

# SWEEP rule on the KD-Toric code

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## 1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex  $v$  looks for a suggestion  $\hat{v}$  on  $\uparrow(v) \cap lk_2$  that matches  $\sigma_v^+$ . If there are more than one, then pick the one which has the minimal weight.

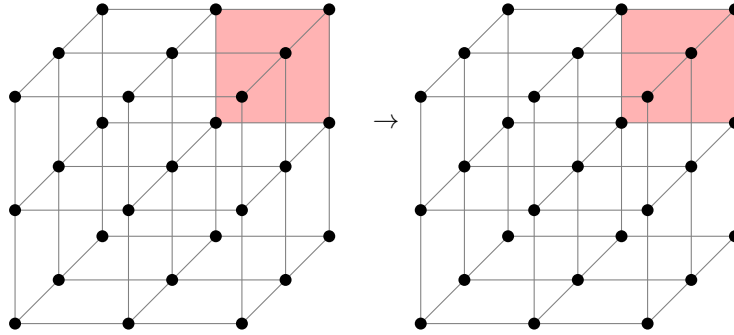


Figure 1: SWEEP rule

## 2 Span Expansion

Define the potentials functions  $L_i : \mathbb{R}^d \rightarrow \mathbb{R}$  as follow<sup>1</sup>:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d + 1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d + 1 \end{cases}$$

The **Size** of a subset  $S$  is defined as:

$$\text{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let  $e$  be an unsatisfied check at time  $t$ , and let  $H_e$  be the unsatisfied checks at time  $t - 1$ . Then there is a set  $H'_e \subset H_e$  such that:

1. For any  $i < d + 1$  there exists  $e' \in H'_e$  such  $L_i(e') \geq L_i(e)$ .
2. There exists  $e' \in H'_e$  such that  $L_{d+1}(e') \leq L_{d+1}(e) + 1$ .

*Proof.* Consider a vertex  $v$  which at time  $t$  adjoins to an unsatisfied edge  $e$ , let  $u, w$  be vertices which have a positive faces that adjoin to  $e$ .

<sup>1</sup>Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. If  $\hat{u}|_e \neq \emptyset$ , then the parallel edge to  $e$  going out from  $u$  belongs to  $H_e$ . Denote it by  $e_u$ , taking  $e_u$  satisfies at least two (three in 4D) of the first  $d$  inequalities and the last  $d + 1$ th inequality. Denote by  $x \in [d]$  the index of the inequality which isn't satisfied by  $e_u$ . So it remains to prove the existence of  $e' \in H_e$  for which  $L_x(e') \geq L_x(e)$ .

Consider the assignment of bits at time  $t - 1$  over the complex, and consider the graph  $G = (\mathcal{F}, E)$  induced by associating vertices with any face that is set to 1 by the assignment. Two vertices are connected if their origin faces share an edge. Let  $K \subset \mathcal{F}$  be a connected component of faces that contains the faces supported on  $u$ .

We say that  $K$  is  $d$  far from  $u$  if:

$$\max_{f \in K} |f_x - u_x| \leq d$$

Now, we will prove by induction on  $d$  that there is a face  $f \in K$  with  $f_x \geq v_x$  such one of its edge is not satisfied.

- (a) **Base.** The edge from  $v$  which point at the  $\hat{z}$  direction must not be satisfied.

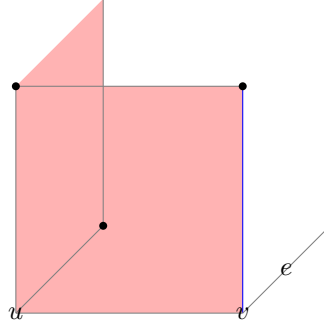


Figure 2: The induction base: In red, we have the faces on the support of  $u$ . The blue edge has the same  $x$ -coordinate as  $e$ . Any attempt to zero out the syndrome on the edge results in adding a face that is at least 2-far from  $u$ . Any configuration which is at most 1-far from  $u$  has to have that local view, yet might be connected to more faces which don't adjoin the blue edge.

- (b) **Assumption.** Assume that for any  $K$  which is at most  $d - 1$  far from  $u$  there is a face, with adjoins edge that is both has a  $x$ -coordinate that is greater than  $u_x$  and is not satisfied.
- (c) **Induction Step.** Consider a connected component  $K$  that is  $d$ -far from  $u$ . Notice that if by substitute a co-boundary (stabilizer) from  $K$  we get  $K'$  that is at most  $d'$  far from  $u$  for some  $d' < d$  then we done. That because:  $\partial_2 K' = \partial_2(K + z) = \partial_2 K$  for some  $z \in \ker \partial_2$ .

On the other hand, [\[COMMENT\]](#) Finish that.

2. Else, if for any  $u \in \downarrow(u) \cap lk_1$  it holds that  $\hat{u}|_e = \emptyset$ , then it must be that  $e \in H_e$  because  $\hat{v}$  can only decrease the syndrome on  $v$ . Moreover, it follows that  $\hat{v} = \emptyset$  (otherwise, the suggestion would zero out the syndrome on  $\uparrow(v) \cap lk_2$ ). Thus,  $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$  there is also an edge  $e' \in H_e$  where  $e' \in \downarrow(v)$ . In that case, take  $e \in H_e$  as the edge satisfies the first  $d$  equalities and  $e'$  for satisfying the  $d + 1$ th inequality.

□

### 3 Investigation

[\[COMMENT\]](#) To do: understands how the investigation should look like. Plot an example to such tree. The model goes as follow, at each even iteration we toss a noise, then in each odd iteration we perform a single SWEEP rule iteration. That allows to track over the history of the errors.

We say that a check  $e \in \text{Error}$ , if there is a face  $f$  that adjoin to  $e$  and belongs to the Error.

**Definition 3.1** (Investigation). For any unsatisfied check  $e$  at time  $t$ , we define  $V$  (the set of checks which participated in 'misleading' of  $e$ ), and two sets of edges on  $V$ , Arrows and Forks via the following algorithm:

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1 let  $V = \{e\}$ , Arrows =  $\emptyset$ , Forks =  $\emptyset$ 
2 let  $Q \leftarrow \text{queue } \{e\}$ 
3 while  $Q$  is not empty do
4    $e' \leftarrow \text{dequeue } Q$ 
5   if  $e' \notin \text{Error}$  then
6      $e_1, \dots, e_l \leftarrow \text{Excuse}(e')$ 
7     Insert  $\{e_i, e'\}$  to Arrows
8     Insert  $\{e_i, e_j\}$  to Forks
9   end
10 end

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### Algorithm 1: Investigation

